



Astronauts: A day in their life

How things are done under zero-gravity situations...

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Google Commander Chris Hadfield and you will come across news of his team's recent Earth landing, fascinating space photographs, some quirky videos from inside the International Space Station and so on. We can safely say that no other astronaut has used social media this wisely to create excitement about astronomy, spacecrafts, spacewalks and so on. No wonder he has 960,000 plus followers on Twitter and by the time we go to print, expect that number to cross the million mark.

We all know that astronauts carry out a lot of experiments on board, plus there is maintenance work to be done on the space station – all in zero gravity. While understanding that is a topic we would love to explore sometime in the future, for this month we decided to cover the absolute basics. How do you perform regular everyday tasks in the absence of gravity?

Hygiene

The first thing we do on waking up – at least most of us – is brush our teeth. Some of us do it before sleeping as well. Having a bath is the next logical step. But both these things involve the use of lots of water, something a lot of us take an unnecessary advantage of. But up in space, there is a different way of doing things.

Most of the liquids taken inside a space craft are carried inside pouches which have a straw coming out of them. You squeeze out water from a pouch just like you squeeze out toothpaste from a tube. Since you do not have access to running water, a couple of drops of water cover the toothpaste on your brush. The foam

residue generated due to brushing cannot be spit out as there is no gravity here to drain it. So you either wipe off the foam on a disposable cloth or like most astronauts – swallow it!

Taking a shower is again out of the question, as water will just float in microgravity and won't really fall on your body or drain out. So the next best way to clean up is using a damp cloth. Special kind of rinseless shampoos are employed inside a space craft which will generate very little or no foam. Astronauts can shampoo their hair with their clothes on as the shampoo just tends to stick to their hair. They need to gently massage their hair to clean them and then wipe off the shampoo.

Space Potty

Going to the toilet is one ritual all of us go through on a daily basis, the process helped in major part by gravity. In a microgravity environment, you can just imagine the difficulty in performing this task and of course you just cannot use the same terrestrial implementations to deal with poop and urine. Enter a special Space Commode!



Space Potty

The commode appears different from the ones seen on earth. For starters, they have a base to hold your feet in position and thigh restraints to hold you in the seated position. In case you want to urinate standing up, you have toe-bars under which you slip in your feet. The commode opening is minor when compared to the western style commodes we are used to and there is a separate pipe with a funnel-like implement at its mouth for urination. The commodes have a suction mechanism to suck all the waste material. So flush is basically flowing air (instead of water) which pulls the waste away from the astronaut's body.

You need to make sure that you aim right. Trainee astronauts who are not very sure of their 'alignment', have a 'Positional Trainer' commode, which resembles a real space potty with one minor addition – the trainer has a camera placed within, which is hooked up to a monitor where the trainee can adjust his/her seating alignment! There are a variety of disinfectant wipes available to wipe up once the business is done.

But what happens during a spacewalk? Well, according to Cmdr Hadfield, astronauts on space walks use special 'absorbent undergarments' like an adult diaper, to take care of waste.

The next logical question that may be bothering you is, 'Where is the waste disposed off?' Solid waste is dried and all the moisture is removed, compressed and kept in an on-board storage container. It is then disposed off when the spacecraft lands on earth. As for liquid waste, it is either sent into space or on the International Space Station it is recycled through a special water treatment plant and turned back into drinking water (Eew?!)